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Inventor(s)	Avlani; Shalin et al.

Multi-layer approach to improving generation of field extraction models

Abstract

A computer-implemented process for generating cluster templates used for creating extraction models includes the following operations. A plurality of training files associated with a selected class are received. An automated visual analysis is performed on each of the plurality of training files. An automated contextual analysis is performed on each of the plurality of training files. A first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis is performed. A second clustering of one of the plurality of clusters into a second plurality of clusters is performed using results from the automated contextual analysis. Cluster templates for the first and second plurality of clusters are generated.

Inventors: Avlani; Shalin (San Jose, CA), Desai; Rajesh M. (San Jose, CA), Shah; Mayank Vipin (San Jose, CA), Gao; Xiaoying (San Jose, CA)

Applicant: INTERNATIONAL BUSINESS MACHINES CORPORATION (Armonk, NY)

Family ID: 95253093

Assignee: International Business Machines Corporation (Armonk, NY)

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Primary Examiner: Nguyen; Merilyn P

Attorney, Agent or Firm: Harrity & Harrity, LLP

Background/Summary

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR A JOINT INVENTOR

(1) The following disclosure is submitted under 35 U.S.C. 102(b)(1)(A): DISCLOSURE: "IBM Automation Document Processing," Automation Document Processing for ICP4BA," 2023 Jun. 30.

BACKGROUND

(2) The present invention relates to electronic document processing, and more specifically, to using generating field extraction templates for use in data extraction during the processing of electronic documents.

(3) Form data extraction is a known technology in which data is extracted from fields in documents. Oftentimes, an enterprise may require data to be extracted from a great number of forms that cannot be done manually. Consequently, technology has been created that includes tools and algorithms to automate the process. The collection of data from a single type of document with a known format is a straight-forward process. In this instance, a form extraction model can be used by a computer to extract data from certain locations in a form since the format is known.

(4) However, in many instances, documents of the same type can be formatted differently. For example, the W2 form that is used to report income may vary from company to company and/or from state to state depending upon the reporting requirements. As another example, the electric bills provided by an electricity provider may vary from provider to provider. In these instances, a single form extraction model cannot be used for each of these different documents of the same type. While it is possible to generate form extraction models for each of these different documents of the same type, this process can be cumbersome and time-inefficient.

SUMMARY

(5) A computer-implemented process for generating cluster templates used for creating extraction models includes the following operations. A plurality of training files associated with a selected class are received. An automated visual analysis is performed on each of the plurality of training files. An automated contextual analysis is performed on each of the plurality of training files. A first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis is performed. A second clustering of one of the plurality of clusters into a

second plurality of clusters is performed using results from the automated contextual analysis. Cluster templates for the first and second plurality of clusters are generated.

(6) In other aspects, the process includes receiving annotations for a first training file within a particular one of the first and second plurality of clusters, and the annotations are automatically propagated to each of a plurality of other training files within the particular one of the first and second plurality of clusters. A cluster annotation template is generated for the particular one of the first and second plurality of clusters using the annotations. Also, an instruction to modify the annotations is received from the user, and the cluster annotation template is modified based upon the instruction. Field extraction models are generated for each of the first and second plurality of clusters using the cluster templates and cluster annotation templates. Additionally, the automated visual analysis can include generating an image hash for each of the plurality of training files, and extracting structural information for each of the plurality of training files. The image hash can include generating a shingle, and the automated contextual analysis includes generating a Term Frequency-Inverse Document Frequency vector for each of the plurality of training files. Additionally, the process can include receiving a document for data extraction; identifying a classification of the document; selecting a field extraction model based upon the classification; and extracting data from the document using the field extraction model.

(7) A computer hardware system for generating cluster templates used for creating extraction models includes a hardware processor configured to perform the following executable operations. A plurality of training files associated with a selected class are received. An automated visual analysis is performed on each of the plurality of training files. An automated contextual analysis is performed on each of the plurality of training files. A first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis is performed. A second clustering of one of the plurality of clusters into a second plurality of clusters is performed using results from the automated contextual analysis. Cluster templates for the first and second plurality of clusters are generated.

(8) In other aspects, the system includes receiving annotations for a first training file within a particular one of the first and second plurality of clusters, and the annotations are automatically propagated to each of a plurality of other training files within the particular one of the first and second plurality of clusters. A cluster annotation template is generated for the particular one of the first and second plurality of clusters using the annotations. Also, an instruction to modify the annotations is received from the user, and the cluster annotation template is modified based upon the instruction. Field extraction models are generated for each of the first and second plurality of clusters using the cluster templates and cluster annotation templates. Additionally, the automated visual analysis can include generating an image hash for each of the plurality of training files, and extracting structural information for each of the plurality of training files. The image hash can include generating a shingle, and the automated contextual analysis includes generating a Term Frequency-Inverse Document Frequency vector for each of the plurality of training files. Additionally, the process can include receiving a document for data extraction; identifying a classification of the document; selecting a field extraction model based upon the classification; and extracting data from the document using the field extraction model.

(9) A computer program product includes a computer readable storage medium having stored therein program code for generating cluster templates used for creating extraction models. The program code, which when executed by a computer hardware system, cause the computer hardware system to perform the following. A plurality of training files associated with a selected class are received. An automated visual analysis is performed on each of the plurality of training files. An automated contextual analysis is performed on each of the plurality of training files. A first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis is performed. A second clustering of one of the plurality of clusters into a second plurality of clusters is performed using results from the automated contextual analysis.

Cluster templates for the first and second plurality of clusters are generated.

(10) In other aspects, the computer program product includes receiving annotations for a first training file within a particular one of the first and second plurality of clusters, and the annotations are automatically propagated to each of a plurality of other training files within the particular one of the first and second plurality of clusters. A cluster annotation template is generated for the particular one of the first and second plurality of clusters using the annotations. Also, an instruction to modify the annotations is received from the user, and the cluster annotation template is modified based upon the instruction. Field extraction models are generated for each of the first and second plurality of clusters using the cluster templates and cluster annotation templates. Additionally, the automated visual analysis can include generating an image hash for each of the plurality of training files, and extracting structural information for each of the plurality of training files. The image hash can include generating a shingle, and the automated contextual analysis includes generating a Term Frequency-Inverse Document Frequency vector for each of the plurality of training files. Additionally, the process can include receiving a document for data extraction; identifying a classification of the document; selecting a field extraction model based upon the classification; and extracting data from the document using the field extraction model.

(11) This Summary section is provided merely to introduce certain concepts and not to identify any key or essential features of the claimed subject matter. Other features of the inventive arrangements will be apparent from the accompanying drawings and from the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-B are block diagrams of a field extraction system according to an aspect of the present invention.

(2) FIG. 2 is a block diagram illustrating an example method for generating template clusters using the architecture of FIGS. 1A-B according to an aspect of the present invention.

(3) FIGS. 3A-B illustrate the clustering of documents according to aspects of the present invention.

(4) FIG. 4 is a block diagram illustrating an example method for generating cluster annotation templates using the architecture of FIGS. 1A-B according to aspects of the present invention.

(5) FIG. 5 is a block diagram illustrating an example method for generating field extraction models using the architecture of FIGS. 1A-B according to aspects of the present invention.

(6) FIG. 6 is a block diagram illustrating an example method for extracting data from documents using the architecture of FIGS. 1A-B according to aspects of the present invention.

(7) FIG. 7 is a block diagram illustrating an example of computer hardware system for implementing the field extraction system of FIGS. 1A-B.

DETAILED DESCRIPTION

(8) Reference is made to FIGS. 1A-B, 2 and 4-5, which respectively illustrate an architecture **100** and methodology **200**, **400**, and **500** for generating cluster templates **147** and field extraction models **157** for use in the automated processing of electronic documents **159**. Referring to FIG. 1A, a plurality of users **105A-C** on client devices **110A-C** can interact with a field extraction system **120** via a network/internet **115**. Although not limited in this manner, the field extraction system **120** is configured to perform the following operations. A plurality of training files **125** associated with a selected class are received by the field extraction system. A document classification trainer **140** is configured to perform an automated visual analysis on each of the plurality of training files **125**. An automated contextual analysis is performed on each of the plurality of training files, and a template clustering engine **145** is configured to perform a first clustering of the plurality of training files **125** into a first plurality of clusters using results from the automated visual analysis. A second clustering of one of the plurality of clusters into a second plurality of clusters is performed using

results from the automated contextual analysis. Cluster templates **147** for the first and second plurality of clusters are then generated.

(9) Although illustrated as a singular entity in FIG. **1A**, the field extraction system **120** is not limited in this manner. With reference to FIG. **1B**, the field extraction system **120** can be comprised of multiple components **120A**, **120B**, with each component **120A**, **120B** containing certain functional aspects of the field extraction system **120**. Although two components **120A**, **120B** are illustrated, the field extraction system **120** is not limited in this manner. For example and as described further described with reference to FIG. **7**, some or all of the functional aspects of the field extraction system **120** can be distributed in the cloud. Additionally, while a single database **135** is illustrated, the database **135** can be also distributed in the individual components **120A**, **120B** and/or within the cloud.

(10) FIG. **2** illustrates a two-stage methodology/process **200** for generating cluster templates **147** that are each associated with an individual cluster **305** of the documents within the plurality of training files **125**. In **210**, a document class to be associated with the cluster templates **147** is defined. The document class reflects the type of information to be retrieved from a document. For example, a mortgage company may require income information as part of a loan underwriting process. In this instance, the type of information needed may include income, assets, and debts. As another example, a solar panel installation company may need energy usage information in order to properly determine the size of a solar panel installation for a home. In this instance, information needed may include energy usage (e.g., KWh) and an address associated with the energy usage. Although not limited in this manner, the document class can be defined by a user **105** interacting with the field extraction system **120** using a client device **110**.

(11) In **220**, a plurality of training files **125** associated with the defined document class is uploaded into the field extraction system **120** via, for example, the interface **130**. The plurality of training files **125** is not limited as to a particular source. For example, the plurality of training files **125** could be provided by the client device **110**, can be loaded from a source external to the field extraction system **120**, and/or be provided by a database **135** associated with the field extraction system **120**. For example, the aforementioned solar panel installation company may use, as the plurality of training files **125**, various different electric bills. As another example, the aforementioned tax company may use, as the plurality of training files **125**, different tax documents.

(12) In **230**, an image hash is generated for each of the training files within the plurality of training files **125**. Techniques for generating an image hash are well-known, and the field extraction system **120** is not limited as to a particular approach. For example, MP5 is a known approach for generating an image hash. However, in certain aspects, a document hash of the image of a training file is generated using a shingle algorithm. A shingle algorithm has a benefit of providing similar values when two similar images are being compared. Consequently, this algorithm can be used to determine how close two images are to one another.

(13) In **240**, structural information is extracted from each document in the plurality of training files **125**. Although not limited in this manner, structural information includes number of pages, page size, number of tables, number of paragraphs, number of links, and number of words. As used here, the term “structural information” includes information that is not “contextual information,” and “contextual information” is information associated with a meaning of the words contained within a particular document.

(14) In **250**, the template clustering engine **145** determines, for the plurality of training files **125**, a number N of clusters and assigns each file within the plurality of training files **125** to a particular cluster and associated cluster template. The N number of clusters can be determined automatically or can be provided by the user **105** via the client device **110** as a seed value. There are many types of techniques that can be used to cluster documents into clusters, and the template clustering engine **145** is not limited as to a particular approach. However, in certain aspects, the template cluster

engine **145** generates a dendrogram (e.g., hierarchical tree structure) with the leaf nodes corresponding to the individual documents. Notably, this approach can be performed more efficiently as opposed to other approaches.

(15) Once the N number of clusters have been identified, the files within the plurality of training files **125** are assigned to a particular one of the clusters (i.e., the documents are clustered together). Although not limited in this manner, the clustering of the documents includes generating a distance matrix using the previously-obtained image hash and structural information for each particular file within the plurality of training files **125**. An example of clustering is illustrated in FIGS. 3A-B.

FIG. 3A represents the plurality of training files **125** (e.g., forms) prior to clustering, and FIG. 3A represents the plurality of training files **125** that have been clustered into five different clusters **305**.

(16) In **255**, a first (or next) cluster **305** in the group of clusters is identified for analysis. Next, in **260**, an automated determination is made as to whether the files/documents within a particular selected cluster **305** are, in fact, properly clustered. For example, while two files may look visually similar, the two files may, in fact, be different. In certain aspects, this determination is performed by generating TF-IDF (Term Frequency-Inverse Document Frequency) vector for each file in the cluster **305**. The TF-IDF is context-sensitive, as opposed to the visual hash and structural information, which are not context-sensitive. If the TF-IDF vectors for all of the documents within a particular cluster **305** are within a predefined value, then a determination can be made that the files within the cluster **305** have been properly clustered and does not require refinement. In addition to or alternatively from the automated determination, the cluster **305** can be manually reviewed (e.g., by the user **105**). If the cluster **305** does not require refine, the process **200** proceeds to **290**. Otherwise, the process **200** moves to operation **270**.

(17) If, in **260**, a determination that a context-based measure reveals that the files within the cluster **305** are not properly clustered using the image hash and the structural information, then in **270**, the files within the cluster can be further clustered into additional clusters using a context-based measure along with the image hash and the structural information. In **275**, after new child clusters have been generated within the parent cluster **305**, these child clusters can be (optionally) manually reviewed (e.g., by the user **105**). If a manual review of the child clusters determines that the child clusters have been inaccurately created (or otherwise not acceptable), then the process **200** can return to operation **270** to perform a re-clustering of the files within the parent cluster **305**.

(18) In **280**, once a determination is made that the particular cluster **305** has been properly generated (i.e., either in **260** or **275**), a cluster template **147** associated with the particular cluster **305** can be generated and subsequently stored, for example, in the database **135**. Although not limited in this manner, the cluster template includes metadata such as: a template ID that identifies that particular cluster template along with metadata that can be used to classify a particular file into the cluster. For example, the metadata can include, for example, an image hash value (or range of image hash values), structural information, and/or value (or range of values) associated with the context information associated with the image cluster (e.g., the TF-IDF vector). Information that specifies the particular document class covered by the cluster template **147** can also be included in a cluster template **147**.

(19) In **290**, after the cluster template **147** has been generated/stored, a determination is made whether all of the clusters **305** have been analyzed. If not, in **285**, the process **200** returns to operation **255**. If yes, the process **200** ends and then can proceed to the process **400** illustrated in FIG. 4.

(20) Referring to FIG. 4, once the multi-layer template clustering process **200** of FIG. 2 has been performed and the cluster templates **147** have been generated and stored, a field annotation process **400** is performed for each identified cluster **305**. In **405**, a first (or next) cluster **305** is identified from the plurality of clusters generated during the process of FIG. 2.

(21) In **410**, a first training file within the identified cluster **305** is annotated to identify information to be extracted from the first training file. The processing of annotation training files is known, and

the field annotator **150** is not limited as to any particular approach so capable. Typically, these approaches include visually presenting the training file to the user **105** via the client device **110** and receiving, via the client device **110**, a bounding box(es) that encompass one or more areas in the training file that contain the information to be extracted. These bounding boxes are computer data structures that will then be associated with the particular training file. In certain instances, the training file being visually presented to the user **105** via the client device **110** can already include annotations. If so, the user can choose, via the client device **110**, to either accept or reject these already-included annotations.

(22) In **415**, once the one or more annotations have been generated/approved for the first training file within the identified cluster **305**, the field annotator **150** is configured to generate a cluster annotation template that is specific to the identified cluster **305**. Specifically, the cluster annotation template describes how the one or more annotations generated/approved for the first training file can be applied to other training files within the identified cluster **305**.

(23) In **420**, the annotations to the first training file are propagated to the other training files in the same cluster **305** as the first training file. This can be performed, for example, by automatically applying, using the field annotator **150**, the cluster annotation template to each of the other training files in the cluster **305**.

(24) In **430**, once the annotations have been propagated to the other training files in the cluster **305**, the user **105** can then review each of the other training files to determine whether the propagated annotations are accurate. In **440**, if the annotations (e.g., the bounding box(es)) are accurate, the process **400** proceeds to **443**, to determine if all of the other training files in the particular cluster **305** have been reviewed. If not, the process **400** loops back to continue review of the other training files within the cluster **305**.

(25) If, in **443**, a determination is made that the annotations for all of the training files in the particular cluster **305** have been reviewed, the process **400** proceeds to **447**, which determines whether all clusters have been analyzed. In **447**, if all the clusters **305** have been analyzed, then the process **400** ends in **449**. Otherwise, the process returns to **405** to select a next cluster to be analyzed in **410**.

(26) If the propagated annotation is not correct, in **450**, the user **105** can select to refine the propagated annotation for a particular file within the cluster **305**. This process can be the same or similar to the operations performed in **410**. Depending upon the amount of changes made, in **460**, a manual or automatic determination can be made as to whether to refine the cluster annotation template associated with the cluster **305**. For example, if the number and/or extent of the changes are minor, a determination can be made not to modify the annotation. However, if, for example, the changes are substantial, the cluster annotation template associated with the cluster **305** can be modified in **470**. The process can then proceed to **443** to determine if all of the training files in the cluster have been reviewed.

(27) In **480**, depending, for example, upon the number of changes needed to be made, a determination can be made as to whether to recluster the training files within the cluster **305**. In this instance, for example, the need to correct the propagated annotation may be because the training files in the cluster **305** were not clustered properly. If this determination is made, then the files can be reclustered in **490**. This can include, for example, providing automated feedback that can be used to recluster as in operations **250** or **270**.

(28) If, in **460**, a determination is made to refine the cluster annotation template, then the cluster annotation template is refined in **470** based upon the refinement of the annotations in **450**. Additionally, in **475**, the one or more annotations indicated by the revised cluster annotation template can be automatically propagated to other training files within the cluster **305** in a manner similar to that described in **420**.

(29) Referring to FIG. 5, a process of generating field extraction models **157** is disclosed. Document classification results **510** and field annotation results **520** are fed into the field extraction

trainer **155**. The classification of documents in **510** is a known process that is able to classify a particular document into a particular document class. The field annotation information includes the specific type of information that is to be extracted from a particular document. Accordingly, the results **510**, **520** include template ID, image hash, and the defined field annotation information. In **530**, using these results **510**, **520**, the field extraction trainer **155** generates for each particular cluster template **147** a field extraction model **157** that is specific to the particular cluster template **147**. For example, the image hash, when saved as part of the field extraction model **157**, can be used to better identify what field extraction model **157** should be used for documents that match the image hash. In **540**, the generated field extraction models **157** can be saved into the database **135**.

(30) Referring to FIG. **6**, a process **600** for classifying a particular document **159** is disclosed. In **610**, the document **159** is received by an interface **160** of the field extraction system **120**. Once the document **159** has been received, a classifier **170** is used to classifying the document into a document type in **620**. For example, the classifier **170** may identify the document **159** as a W2 form, a tax form, or an electric bill. Next, in **630** and based upon the document type in **160**, the classifier **170** determines a cluster, within the classification, associated with the document **159** based upon information contained within the cluster templates **147**.

(31) After the document **159** has been classified and a particular cluster template **147** has been identified for use with the document **159**, in **640**, a field extraction model **157** can be identified for use with the document **159**. The field extraction model **157** can be determined based upon, for example, the template ID, an image hash of the document **159**, structural information of the document **159**, and/or text within the document **159**. For example, an image hash of the document **159** (e.g., using the processing described with reference to operation **230** in FIG. **2**) can be generated and that image hash of the document **159** can be compared to image hashes associated with each of the field extraction models **157**.

(32) If a correct field extraction model **157** is identified in **670**, data **195** is extracted from the document **159** using the field extraction model **157**, and that data **195** can be subsequently stored in the database **135**. Alternatively, in **660**, if an incorrect field extraction model **157** was initially identified in **640**, a different field extraction model **157** can be selected to extract data **195** from the document **159**.

(33) As defined herein, the term “responsive to” means responding or reacting readily to an action or event. Thus, if a second action is performed “responsive to” a first action, there is a causal relationship between an occurrence of the first action and an occurrence of the second action, and the term “responsive to” indicates such causal relationship.

(34) As defined herein, the term “real time” means a level of processing responsiveness that a user or system senses as sufficiently immediate for a particular process or determination to be made, or that enables the processor to keep up with some external process.

(35) As defined herein, the term “automatically” means without user intervention.

(36) Referring to FIG. **7**, computing environment **700** contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as code block **750** for implementing the operations of the field extraction system **120**.

Computing environment **700** includes, for example, computer **701**, wide area network (WAN) **702**, end user device (EUD) **703**, remote server **704**, public cloud **705**, and private cloud **706**. In certain aspects, computer **701** includes processor set **710** (including processing circuitry **720** and cache **721**), communication fabric **711**, volatile memory **712**, persistent storage **713** (including operating system **722** and method code block **750**), peripheral device set **714** (including user interface (UI), device set **723**, storage **724**, and Internet of Things (IoT) sensor set **725**), and network module **715**. Remote server **704** includes remote database **730**. Public cloud **705** includes gateway **740**, cloud orchestration module **741**, host physical machine set **742**, virtual machine set **743**, and container set **744**.

(37) Computer **701** may take the form of a desktop computer, laptop computer, tablet computer,

smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database **730**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. However, to simplify this presentation of computing environment **700**, detailed discussion is focused on a single computer, specifically computer **701**. Computer **701** may or may not be located in a cloud, even though it is not shown in a cloud in FIG. 7 except to any extent as may be affirmatively indicated.

(38) Processor set **710** includes one, or more, computer processors of any type now known or to be developed in the future. As defined herein, the term “processor” means at least one hardware circuit (e.g., an integrated circuit) configured to carry out instructions contained in program code. Examples of a processor include, but are not limited to, a central processing unit (CPU), an array processor, a vector processor, a digital signal processor (DSP), a field-programmable gate array (FPGA), a programmable logic array (PLA), an application specific integrated circuit (ASIC), programmable logic circuitry, and a controller. Processing circuitry **720** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **720** may implement multiple processor threads and/or multiple processor cores. Cache **721** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **710**. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In certain computing environments, processor set **710** may be designed for working with qubits and performing quantum computing.

(39) Computer readable program instructions are typically loaded onto computer **701** to cause a series of operational steps to be performed by processor set **710** of computer **701** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods discussed above in this document (collectively referred to as “the inventive methods”). These computer readable program instructions are stored in various types of computer readable storage media, such as cache **721** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **710** to control and direct performance of the inventive methods. In computing environment **700**, at least some of the instructions for performing the inventive methods may be stored in code block **750** in persistent storage **713**.

(40) A computer program product embodiment (“CPP embodiment” or “CPP”) is a term used in the present disclosure to describe any set of one, or more, storage media (also called “mediums”) collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A “storage device” is any tangible, hardware device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing.

(41) A computer readable storage medium, as that term is used in the present disclosure, is not to be

construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

(42) Communication fabric **711** is the signal conduction paths that allow the various components of computer **701** to communicate with each other. Typically, this communication fabric **711** is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used for the communication fabric **711**, such as fiber optic communication paths and/or wireless communication paths.

(43) Volatile memory **712** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, the volatile memory **712** is characterized by random access, but this is not required unless affirmatively indicated. In computer **701**, the volatile memory **712** is located in a single package and is internal to computer **701**. In addition to alternatively, the volatile memory **712** may be distributed over multiple packages and/or located externally with respect to computer **701**.

(44) Persistent storage **713** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of the persistent storage **713** means that the stored data is maintained regardless of whether power is being supplied to computer **701** and/or directly to persistent storage **713**. Persistent storage **713** may be a read only memory (ROM), but typically at least a portion of the persistent storage **713** allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage **713** include magnetic disks and solid state storage devices. Operating system **722** may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface type operating systems that employ a kernel. The code included in code block **750** typically includes at least some of the computer code involved in performing the inventive methods.

(45) Peripheral device set **714** includes the set of peripheral devices for computer **701**. Data communication connections between the peripheral devices and the other components of computer **701** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet.

(46) In various aspects, UI device set **723** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **724** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **724** may be persistent and/or volatile. In some aspects, storage **724** may take the form of a quantum computing storage device for storing data in the form of qubits. In aspects where computer **701** is required to have a large amount of storage (for example, where computer **701** locally stores and manages a large database) then this storage **724** may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. Internet-of-Things (IoT) sensor set **725** is made up of sensors that can be used in IoT applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

(47) Network module **715** is the collection of computer software, hardware, and firmware that allows computer **701** to communicate with other computers through a Wide Area Network (WAN)

702. Network module **715** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In certain aspects, network control functions and network forwarding functions of network module **715** are performed on the same physical hardware device. In other aspects (for example, aspects that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **715** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **701** from an external computer or external storage device through a network adapter card or network interface included in network module **715**.

(48) WAN **702** is any Wide Area Network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some aspects, the WAN **702** may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN **702** and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

(49) End user device (EUD) **703** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **701**), and may take any of the forms discussed above in connection with computer **701**. EUD **703** typically receives helpful and useful data from the operations of computer **701**. For example, in a hypothetical case where computer **701** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **715** of computer **701** through WAN **702** to EUD **703**. In this way, EUD **703** can display, or otherwise present, the recommendation to an end user. In certain aspects, EUD **703** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

(50) As defined herein, the term “client device” means a data processing system that requests shared services from a server, and with which a user directly interacts. Examples of a client device include, but are not limited to, a workstation, a desktop computer, a computer terminal, a mobile computer, a laptop computer, a netbook computer, a tablet computer, a smart phone, a personal digital assistant, a smart watch, smart glasses, a gaming device, a set-top box, a smart television and the like. Network infrastructure, such as routers, firewalls, switches, access points and the like, are not client devices as the term “client device” is defined herein. As defined herein, the term “user” means a person (i.e., a human being).

(51) Remote server **704** is any computer system that serves at least some data and/or functionality to computer **701**. Remote server **704** may be controlled and used by the same entity that operates computer **701**. Remote server **704** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **701**. For example, in a hypothetical case where computer **701** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **701** from remote database **730** of remote server **704**. As defined herein, the term “server” means a data processing system configured to share services with one or more other data processing systems.

(52) Public cloud **705** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **705** is performed by the computer hardware and/or software of cloud orchestration module **741**. The computing resources provided by public cloud **705** are typically implemented by virtual computing

environments that run on various computers making up the computers of host physical machine set **742**, which is the universe of physical computers in and/or available to public cloud **705**.

(53) The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **743** and/or containers from container set **744**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **741** manages the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **740** is the collection of computer software, hardware, and firmware that allows public cloud **705** to communicate through WAN **702**.

(54) VCEs can be stored as “images,” and a new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

(55) Private cloud **706** is similar to public cloud **705**, except that the computing resources are only available for use by a single enterprise. While private cloud **706** is depicted as being in communication with WAN **702**, in other aspects, a private cloud **706** may be disconnected from the internet entirely (e.g., WAN **702**) and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this aspect, public cloud **705** and private cloud **706** are both part of a larger hybrid cloud.

(56) Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

(57) As another example, two blocks shown in succession may, in fact, be accomplished as one step, executed concurrently, substantially concurrently, in a partially or wholly temporally overlapping manner, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions. Each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s).

(58) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or

“comprising,” when used in this disclosure, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. (59) Reference throughout this disclosure to “one embodiment,” “an embodiment,” “one arrangement,” “an arrangement,” “one aspect,” “an aspect,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment described within this disclosure. Thus, appearances of the phrases “one embodiment,” “an embodiment,” “one arrangement,” “an arrangement,” “one aspect,” “an aspect,” and similar language throughout this disclosure may, but do not necessarily, all refer to the same embodiment.

(60) The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The term “coupled,” as used herein, is defined as connected, whether directly without any intervening elements or indirectly with one or more intervening elements, unless otherwise indicated. Two elements also can be coupled mechanically, electrically, or communicatively linked through a communication channel, pathway, network, or system. The term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will also be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms, as these terms are only used to distinguish one element from another unless stated otherwise or the context indicates otherwise.

(61) The term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may be construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context. As used herein, the terms “if,” “when,” “upon,” “in response to,” and the like are not to be construed as indicating a particular operation is optional. Rather, use of these terms indicate that a particular operation is conditional. For example and by way of a hypothetical, the language of “performing operation A upon B” does not indicate that operation A is optional. Rather, this language indicates that operation A is conditioned upon B occurring.

(62) The foregoing description is just an example of embodiments of the invention, and variations and substitutions. While the disclosure concludes with claims defining novel features, it is believed that the various features described herein will be better understood from a consideration of the description in conjunction with the drawings. The process(es), machine(s), manufacture(s) and any variations thereof described within this disclosure are provided for purposes of illustration. Any specific structural and functional details described are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the features described in virtually any appropriately detailed structure. Further, the terms and phrases used within this disclosure are not intended to be limiting, but rather to provide an understandable description of the features described.

Claims

1. A computer-implemented method for generating cluster templates used for creating extraction models, comprising: receiving a plurality of training files associated with a selected class; performing an automated visual analysis on each of the plurality of training files; performing an automated contextual analysis on each of the plurality of training files; performing a first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis; performing a second clustering of one of the first plurality of clusters into a second plurality of clusters using results from the automated contextual analysis; and generating cluster

templates for the first and second plurality of clusters, wherein the first and the second plurality of clusters are clusters of the plurality of training files.

2. The method of claim 1, further comprising: receiving, for a first training file within a particular one of the first and second plurality of clusters, annotations; automatically propagating the annotations to each of a plurality of other training files within the particular one of the first and second plurality of clusters; and generating a cluster annotation template for the particular one of the first and second plurality of clusters using the annotations.

3. The method of claim 2, further comprising: receiving, from a user, an instruction to modify the annotations; and modifying the cluster annotation template based upon the instruction.

4. The method of claim 2, further comprising: generating field extraction models for each of the first and second plurality of clusters using the cluster templates and cluster annotation templates.

5. The method of claim 1, wherein the automated visual analysis includes: generating an image hash for each of the plurality of training files, and extracting structural information for each of the plurality of training files.

6. The method of claim 5, wherein the image hash includes generating a shingle.

7. The method of claim 1, wherein the automated contextual analysis includes generating a Term Frequency-Inverse Document Frequency vector for each of the plurality of training files.

8. The method of claim 1, further comprising: receiving a document for data extraction; identifying a classification of the document; selecting a field extraction model based upon the classification; and extracting data from the document using the field extraction model.

9. A computer hardware system for generating cluster templates used for creating extraction models, comprising: a hardware processor configured to perform the following executable operations: receiving a plurality of training files associated with a selected class; performing an automated visual analysis on each of the plurality of training files; performing an automated contextual analysis on each of the plurality of training files; performing a first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis; performing a second clustering of one of the first plurality of clusters into a second plurality of clusters using results from the automated contextual analysis; and generating the cluster templates for the first and second plurality of clusters, wherein the first and the second plurality of clusters are clusters of the plurality of training files.

10. The system of claim 9, wherein the hardware processor is further configured to perform the following: receiving, for a first training file within a particular one of the first and second plurality of clusters, annotations; automatically propagating the annotations to each of a plurality of other training files within the particular one of the first and second plurality of clusters; and generating a cluster annotation template for the particular one of the first and second plurality of clusters using the annotations.

11. The system of claim 10, wherein the hardware processor is further configured to perform the following: receiving, from a user, an instruction to modify the annotations; and modifying the cluster annotation template based upon the instruction.

12. The system of claim 10, wherein the hardware processor is further configured to perform the following: generating field extraction models for each of the first and second plurality of clusters using the cluster templates and cluster annotation templates.

13. The system of claim 9, wherein the automated visual analysis includes: generating an image hash for each of the plurality of training files, and extracting structural information for each of the plurality of training files.

14. The system of claim 13, wherein the image hash includes generating a shingle.

15. The system of claim 9, wherein the automated contextual analysis includes generating a Term Frequency-Inverse Document Frequency vector for each of the plurality of training files.

16. The system of claim 9, wherein the hardware processor is further configured to perform the following: receiving a document for data extraction; identifying a classification of the document;

selecting a field extraction model based upon the classification; and extracting data from the document using the field extraction model.

17. A computer program product, comprising: a computer readable storage medium having stored therein program code for generating cluster templates used for creating extraction models, the program code, which when executed by a computer hardware system, cause the computer hardware system to perform: receiving a plurality of training files associated with a selected class; performing an automated visual analysis on each of the plurality of training files; performing an automated contextual analysis on each of the plurality of training files; performing a first clustering of the plurality of training files into a first plurality of clusters using results from the automated visual analysis; performing a second clustering of one of the first plurality of clusters into a second plurality of clusters using results from the automated contextual analysis; and generating the cluster templates for the first and second plurality of clusters, wherein the first and the second plurality of clusters are clusters of the plurality of training files.

18. The computer program product of claim 17, wherein the computer hardware system is further configured to perform the following: receiving, for a first training file within a particular one of the first and second plurality of clusters, annotations; automatically propagating the annotations to each of a plurality of other training files within the particular one of the first and second plurality of clusters; generating a cluster annotation template for the particular one of the first and second plurality of clusters using the annotations; receiving, from a user, an instruction to modify the annotations; modifying the cluster annotation template based upon the instruction; and generating field extraction models for each of the first and second plurality of clusters using the cluster templates and cluster annotation templates.

19. The computer program product of claim 17, wherein the automated visual analysis includes: generating an image hash for each of the plurality of training files, and extracting structural information for each of the plurality of training files, the image hash includes generating a shingle, and the automated contextual analysis includes generating a Term Frequency-Inverse Document Frequency vector for each of the plurality of training files.

20. The computer program product of claim 17, wherein the program code further causes the computer hardware system to perform: receiving a document for data extraction; identifying a classification of the document; selecting a field extraction model based upon the classification; and extracting data from the document using the field extraction model.
